Algebra 1 – Transformations Worksheet

Visualizing Transformations Through Motion, Springs, and Mirrors

Name: _____ **Date:** _____

Learning Goals

By the end of this worksheet, students should be able to:

- Translate points on the coordinate plane
- Understand horizontal and vertical stretch/shrink visually
- Identify how coordinates change during transformations
- Reflect figures across lines
- Explain transformations conceptually

Transformation Vocabulary

- **Translation** = slide
- **Reflection** = flip like a mirror
- **Stretch** = move farther from an axis
- **Shrink** = move closer to an axis

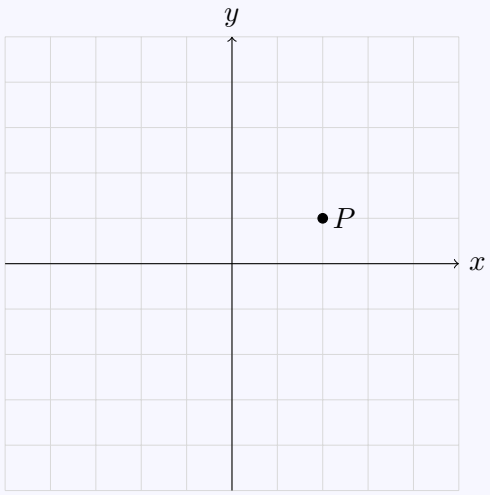
A. Warm Up – Directions from a Point

Start from the point P. Find the new point after moving in the given direction.

$$\text{Left } 1 \to (x-1, y) \qquad \text{Right } 2 \to (x+2, y)$$

$$\text{Up } 3 \to (x, y+3) \qquad \text{Down } 4 \to (x, y-4)$$

Example: $P(2, 1)$



- Left 1: _____
- Right 2: _____
- Up 3: _____
- Down 4: _____

Key Idea: Moving left/right changes the x-coordinate. Moving up/down changes the y-coordinate.

B. Stretch / Shrink (The Spring Model)

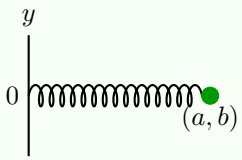
Think of a spring attached to an axis.

- Horizontal stretch/shrink: spring attached to the **y-axis**
- Vertical stretch/shrink: spring attached to the **x-axis**

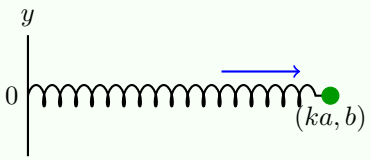
Horizontal Stretch & Shrink

$$(x, y) \to (kx, y)$$

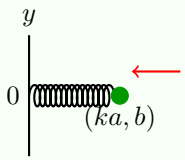
Original $(k = 1)$



Stretch $(k > 1)$



Shrink $(0 < k < 1)$



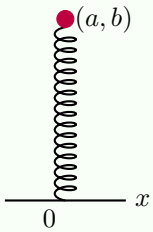
Key Idea: Horizontal stretch changes the distance from the y-axis. The y-coordinate stays the same.

Think About It:
Why does the y-coordinate stay the same?

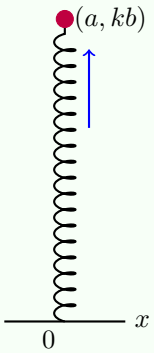
Vertical Stretch & Shrink

$$(x, y) \to (x, ky)$$

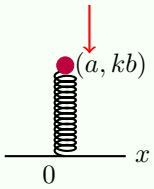
Original $(k = 1)$



Stretch $(k > 1)$



Shrink $(0 < k < 1)$



Key Idea: Vertical stretch changes the distance from the x-axis. The x-coordinate stays the same.

Think About It:
Why does the x-coordinate stay the same?

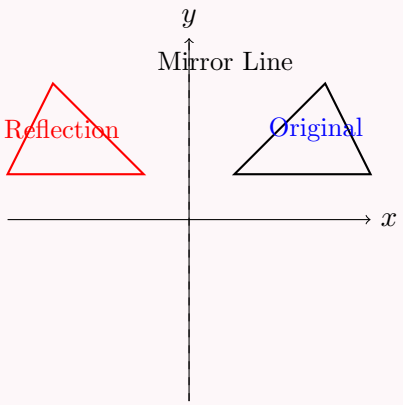
C. Reflections

Over x-axis: $(x, y) \to (x, -y)$

Over y-axis: $(x, y) \to (-x, y)$

Over $y = x$: $(x, y) \to (y, x)$

Over $y = -x$: $(x, y) \to (-y, -x)$



Key Idea: Reflections flip a figure across a line like a mirror.

D. Connection to Functions

Transformations also appear in functions.

$$f(x) + 3$$

moves a graph up 3 units.

$$-f(x)$$

reflects a graph over the x-axis.

$$f\left(\frac{x}{2}\right)$$

creates a horizontal stretch.

$$2f(x)$$

creates a vertical stretch.

Self Reflection

- Which transformation was easiest for you?
- Which transformation was hardest?
- What does "stretch" really mean geometrically?
- What does "reflection" really mean visually?
- What is one mistake you should avoid next time?

Remember: Transformations change position, size, or direction!